

PINE RIVER, MN, USA

WMO: 727508

Lat: 46.725N

Lon: 94.382W

Elev: 395

StdP: 96.67

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 54915

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.0	-27.0	-32.8	0.2	-29.8	-29.9	0.2	-25.6	8.4	-9.7	7.7	-9.2	0.6	300	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	30.9	22.6	28.7	21.4	27.4	20.3	24.2	28.6	23.0	26.9	21.7	25.4	4.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.3	26.6	22.0	17.5	26.2	20.9	16.3	24.9	75.2	27.7	70.3	26.8	65.2	25.8	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.5	6.7	DB	-34.2	32.8	3.4	1.9	-36.7	34.2	-38.6	35.3	-40.5	36.4	-43.0	37.8	(4)
(5)			WB	-34.6	25.6	3.3	1.5	-37.0	26.7	-39.0	27.5	-40.9	28.4	-43.3	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-12.1	-10.6	-2.4	5.2	12.0	17.2	20.2	18.2	13.9	6.4	-0.8	-9.3	(6)	
	DBStd	12.56	7.78	7.22	7.14	5.46	4.56	3.59	3.27	3.28	4.72	4.88	5.93	7.19	(7)	
	HDD10.0	2911	686	576	388	159	30	1	0	0	15	133	325	599	(8)	
	HDD18.3	5062	944	809	644	393	203	61	18	42	147	369	574	857	(9)	
	CDD10.0	1056	0	0	2	17	92	218	316	256	132	23	1	0	(10)	
	CDD18.3	166	0	0	0	1	7	28	76	39	14	1	0	0	(11)	
	CDH23.3	1744	0	0	0	10	110	298	789	393	132	12	0	0	(12)	
	CDH26.7	441	0	0	0	2	26	63	229	92	27	2	0	0	(13)	
(14) Wind	WSAvg	2.6	2.5	2.5	2.8	3.2	3.1	2.6	2.2	2.1	2.5	2.7	2.7	2.5	(14)	
(15) Precipitation	PrecAvg	710	18	16	34	54	81	109	104	79	80	79	33	22	(15)	
	PrecMax	855	49	49	89	131	192	190	193	160	180	152	104	54	(16)	
	PrecMin	529	2	1	9	21	31	37	48	22	19	10	4	7	(17)	
	PrecStd	85	13	11	22	25	37	40	40	39	34	40	24	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	8.0	18.7	24.1	29.7	31.1	32.9	32.3	29.8	25.0	17.5	6.4	(19)	
		MCWB	3.7	4.9	13.7	13.7	18.9	23.2	24.0	23.4	21.2	16.3	12.1	3.8	(20)	
	2%	DB	2.5	5.0	13.9	21.1	26.4	28.0	31.2	28.9	27.2	20.9	13.0	3.7	(21)	
		MCWB	0.8	2.2	8.8	11.8	16.4	20.8	23.2	21.7	20.0	14.7	9.0	2.2	(22)	
	5%	DB	0.9	2.5	10.2	17.8	23.6	27.0	28.9	27.4	24.1	17.4	11.0	2.1	(23)	
		MCWB	0.0	0.8	6.9	10.0	15.1	19.5	22.0	20.5	18.5	12.4	7.0	1.1	(24)	
	10%	DB	-1.3	0.2	7.4	15.3	21.3	24.8	27.5	26.0	22.4	14.1	7.7	0.3	(25)	
		MCWB	-1.7	-0.8	4.6	8.1	14.1	18.0	21.0	19.6	17.6	10.4	5.3	-0.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	5.7	14.1	15.7	21.3	24.5	26.2	25.5	23.3	17.9	12.9	4.2	(27)	
		MCDB	6.6	8.2	16.1	23.0	26.9	28.9	29.8	29.8	27.4	21.2	15.4	5.5	(28)	
	2%	WB	1.1	2.8	9.8	12.9	18.7	22.8	24.7	23.2	21.5	15.4	10.0	2.4	(29)	
		MCDB	2.3	4.7	13.6	18.6	23.6	26.9	29.1	27.0	25.0	19.8	13.5	3.6	(30)	
	5%	WB	-0.3	0.9	7.1	10.9	16.9	21.0	23.6	21.8	19.9	13.1	7.5	1.1	(31)	
		MCDB	0.6	2.5	9.9	16.9	21.4	24.8	27.7	25.5	23.0	16.6	9.4	1.8	(32)	
	10%	WB	-2.1	-1.0	4.6	9.1	15.1	19.4	22.2	20.6	18.4	11.3	5.4	-0.4	(33)	
		MCDB	-1.7	0.1	7.4	14.6	19.5	23.2	25.9	24.2	21.4	14.0	8.1	0.3	(34)	
(35) Mean Daily Temperature Range	MDBR		10.7	11.5	11.1	11.9	12.6	11.7	12.2	12.2	12.0	10.2	9.2	8.8	(35)	
	5% DB	MCDBR	10.6	12.1	14.7	17.3	17.3	13.9	14.2	14.0	14.7	15.3	14.2	10.5	(36)	
		MCWBR	9.1	9.5	9.7	9.7	9.1	7.4	7.3	7.6	8.4	9.4	9.5	8.8	(37)	
	5% WB	MCDBR	10.2	11.1	14.1	15.3	13.5	11.6	12.6	12.0	11.6	12.9	12.3	9.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.256	0.288	0.308	0.356	0.374	0.375	0.379	0.385	0.354	0.311	0.283	0.263	(40)	
	taud		2.374	2.285	2.386	2.348	2.354	2.382	2.388	2.387	2.453	2.564	2.517	2.398	(41)	
	Ebn at Noon		852	883	920	900	891	890	880	860	860	858	819	799	(42)	
	Edh at Noon		70	95	101	116	119	117	115	110	94	73	62	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.59	2.72	3.83	4.82	5.36	5.86	6.23	5.24	3.91	2.40	1.47	1.10	(44)	
	RadStd		0.18	0.27	0.38	0.41	0.38	0.40	0.27	0.44	0.29	0.37	0.15	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-2.52	N/A	-1.13	-1.26	-1.22	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page